Birla Institute of Technology and Science, Pilani – KK Birla Goa Campus

EEE/ECE/INSTR F241

Microprocessor Programming and Interfacing

Time: 15 minutes **Quiz 2** **Max marks: 10**

Answers should be within the designated boxes
Answers not written with blue/blank pen and overwritten answers will be graded to 0

1. For an 8086-based computer total memory access time is 0.4μs. The processor runs at 8 MHz clock frequency. Ignoring all setup and hold time requirements, the minimum number of wait states required for the proper operation of the system will be

2. Which addressing mode generates the largest machine code for 8086?

3. For 8086, what is the minimum number of T-states required for an instruction fetch machine cycle?

4. What is the machine code in hexadecimal code for 8086 instruction **`MOV AX,CS`**?

5. An 8086 processor is interfaced with a single ROM memory bank with 64 KB capacity. From a practical perspective, what starting physical address you will assign to the ROM (in hex) if no memory address warp around is allowed.

6. In an 8086 processor, if the instruction **`MOV AX,BX`** is stored in the memory location CS:100H, the content of the IP after executing this instruction will be

7. An 8088 processor (it has 8 bit-data bus) based computer requires a total of 1 MB memory. The only discrete SRAM chip available in the market is in 8Kx16 configuration, with WE, RE and CS control signals. Determine the minimum number of discrete SRAM chips that you need to buy to satisfy the system memory requirement

8. What is the assembly equivalent of machine code (in hex) 8A 27?

9. What is the machine code for **`MOV CS,DS`**?

10. An 8088 processor is interfaced with a total of 64 KB RAM (it has 8 bit-data bus). The total memory is composed of 2 discrete RAM chips. If the RAM is mapped to system address space starting from 0x00000 and the partial memory address decoding is done with the help of a single address line, which address line will you choose (Answer in the form of A_0, A_1 etc.)